SOLAR BASED WIRELSS CHARGING SYSTEM

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Introduction

The increasing dependence on mobile devices has created a growing need for reliable and easily accessible charging solutions, particularly in outdoor and off-grid environments where conventional power sources are limited. Most existing charging systems rely heavily on grid electricity, which restricts their usability in remote or rural areas and increases dependency on non-renewable energy sources. To address this challenge, this project proposes a solar-powered mobile charging system that utilizes renewable energy for sustainable operation. By integrating both wired (USB) and wireless (Qi-based) charging technologies, the system offers flexibility, convenience, and improved accessibility, making it a practical and efficient solution for real-world applications.

Aim, Objectives, Motivation

Aim: To develop an efficient solar-powered mobile charging system capable of supporting both wired and wireless charging..

Objectives

- To design and develop a solar-powered mobile charging system using renewable energy.
- To implement hybrid charging by integrating Qi-based wireless and USB wired charging.
- To regulate and manage power flow using a charge controller and battery storage system.
- To evaluate system performance in terms of efficiency, reliability, and practical usability.

Motivation

To reduce dependency on conventional electricity and provide accessible charging solutions in outdoor and remote areas.

Similar Works

Existing systems primarily focus on solar-based wired charging with limited efficiency and lack of advanced features. Many systems do not integrate wireless charging or proper energy management, resulting in reduced usability. This project improves upon existing solutions by combining renewable energy with hybrid charging and optimized system design.



Figure 1

Figure 1

Methodology

The system is divided into three main blocks: energy generation, storage, and utilization. Solar energy is captured using a photovoltaic panel and regulated through a PWM charge controller. The energy is stored in a rechargeable battery and supplied to charging modules through a DC-DC converter. Both wired (USB) and wireless (Qi) charging methods are integrated to provide flexible and efficient power delivery

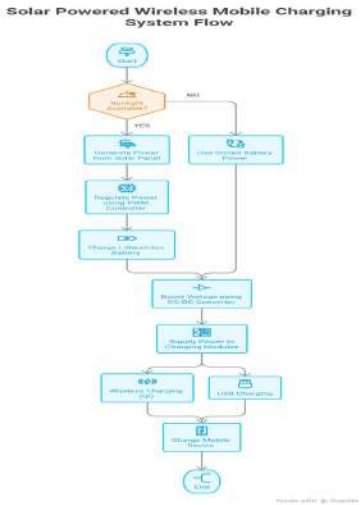


Figure 2

Experimental Setup

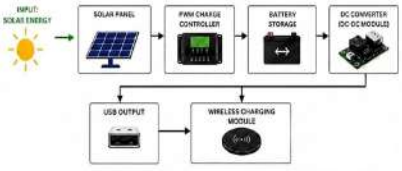
Hardware Components:

- Solar Panel (65W)
- PWM Charge Controller
- Lithium-ion Battery
- DC-DC Boost Converter
- Qi Wireless Charging Module
- USB Charging Module

Working:

- Solar panel converts sunlight into electrical energy
- Energy is regulated using the PWM charge controller
- Battery stores energy for continuous operation
- Buck converter stabilizes output voltage
- Wireless and USB modules deliver power to devices

BLOCK DIAGRAM:



Results

The Solar Powered Wireless Mobile Charging System was tested under real-time conditions and successfully generated, stored, and delivered energy for both wireless and USB charging. The system provided stable output with reliable performance, ensuring continuous operation even under varying sunlight conditions. Both wireless and wired charging modes functioned effectively.

Results

The system achieved an overall efficiency of approximately **45%** under practical conditions. Wireless charging efficiency ranged between **15–25%**, while USB charging efficiency ranged between **25–30%**. The system provided stable voltage output and consistent charging performance, validating its practical applicability.



Figure 5



Figure 6

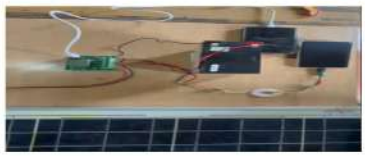


Figure 7

Conclusion and Outcomes of Project

The project successfully demonstrates a solar-powered mobile charging system integrating renewable energy with both wired and wireless charging technologies. The system provides a sustainable and efficient alternative to conventional charging methods. It shows strong potential for real-world deployment and future improvements in efficiency, scalability, and smart energy management.

Outcomes of the Project

- Automated control of classroom devices
- Reduced energy wastage
- IoT and mobile app integration achieved
- Cost-effective and scalable solution
- Scope for future improvements and real-time deployment

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